Clinic Appointment System

Refined User and System Requirements

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User and System Requirements

1. Introduction

Before developing a software system, it is important to understand what the system should do and what users expect from it. Requirements help identify stakeholder needs and define the functions, services, and quality characteristics that the system must provide.

In software engineering, requirements are generally divided into two main categories: user requirements and system requirements.

User Requirements describe what different users and stakeholders need from the system.

System Requirements translate those needs into specific functional and non-functional requirements that guide system development, implementation, and testing.

The Clinic Appointment System aims to simplify appointment scheduling and management by providing a centralized platform for patients, doctors, receptionists, and administrators.

2. User Requirements

User requirements describe the needs and expectations of the individuals who interact with or benefit from the Clinic Appointment System.

2.1 Patient Needs

Patients should be able to:

- Create an account and securely log in to the system.
- Log out of their accounts securely.
- Manage their personal profiles.
- Search for doctors by name and medical specialty.
- View doctor information, including names, specialties, and professional details.
- View doctors' available appointment dates and time slots.
- Book available appointments.
- Receive appointment confirmation after successful booking.
- Cancel appointments according to the clinic's cancellation policy.
- Reschedule appointments based on doctor availability.
- View their upcoming and previous appointments.
- View the current status of their appointments.
- Receive appointment notifications and reminders.
- Access only their own personal and appointment information.

2.2 Doctor Needs

Doctors should be able to:

- Securely log in and log out of the system.
- View and manage their professional profiles.
- Set, update, and manage their appointment availability.
- View their upcoming and previous appointments.

- View relevant patient and appointment details necessary for their work.
- Manage their appointment schedules according to their assigned permissions.
- Access only the information authorized for their role.

2.3 Receptionist Needs

Receptionists should be able to:

- Securely log in and log out of the system.
- Register patients when necessary.
- View and update permitted patient information.
- View doctor information and availability.
- Create appointments on behalf of patients.
- View and manage scheduled appointments.
- Cancel or reschedule appointments according to the clinic's policies and assigned permissions.
- Maintain accurate and organized appointment records.
- Access only the information and functions authorized for their role.

2.4 Administrator Needs

Administrators should be able to:

- Securely log in and log out of the system.
- Manage patient accounts and permitted patient information.
- Create and manage doctor accounts.
- Create and manage receptionist accounts.
- Manage doctor information and professional profiles.
- Manage doctor availability.
- Oversee and manage appointments.
- Manage relevant clinic information and system settings.
- Manage user roles and access permissions.
- Monitor appointment records and overall system activities.

2.5 Clinic Management Needs

Clinic management should be able to:

- Maintain an organized appointment management process.
- Reduce manual appointment scheduling and record-keeping.
- Reduce appointment conflicts and double-booking.
- Maintain accurate and reliable appointment information.
- Ensure efficient coordination between patients, doctors, and clinic staff.
- Monitor appointment activities and overall clinic scheduling operations.
- Improve the efficiency of the clinic's appointment management process.

3. System Requirements

System requirements define the specific functions and quality characteristics that the Clinic Appointment System must provide to satisfy stakeholder needs.

System requirements are divided into functional requirements and non-functional requirements.

3.1 Functional Requirements

Functional requirements describe the specific operations, services, and features that the system must perform.

A. User Account and Authentication

FR-01: The system shall allow patients to register for an account by providing the required registration information.

FR-02: The system shall allow registered users to log in using valid credentials.

FR-03: The system shall provide different access permissions based on the user's assigned role, including patient, doctor, receptionist, and administrator.

FR-04: The system shall allow authenticated users to log out of their accounts.

FR-05: The system shall allow authorized administrators to create and manage doctor and receptionist accounts.

B. Patient Management

FR-06: The system shall allow patients to create and manage their personal profiles.

FR-07: The system shall allow patients to view and update permitted personal information.

FR-08: The system shall allow patients to search for doctors by name and medical specialty.

FR-09: The system shall allow patients to view doctor information, including names, specialties, and professional details.

FR-10: The system shall allow patients to view available appointment dates and time slots for doctors.

C. Appointment Management

FR-11: The system shall allow patients to book available appointments with their selected doctors.

FR-12: The system shall prevent two patients from booking the same doctor's appointment slot, including simultaneous booking attempts.

FR-13: The system shall allow patients to cancel appointments according to the clinic's established cancellation policy.

FR-14: The system shall allow patients to reschedule appointments by selecting another available appointment slot.

FR-15: The system shall release the original appointment slot after a successful cancellation or rescheduling operation.

FR-16: The system shall validate appointment availability before confirming a booking or rescheduling request.

FR-17: The system shall provide appointment confirmation after a successful booking.

FR-18: The system shall include the doctor's name, appointment date, appointment time, and a unique appointment reference in the confirmation details.

FR-19: The system shall allow patients to view their upcoming and previous appointments.

FR-20: The system shall display the current status of appointments, including confirmed, cancelled, and completed appointments.

FR-21: The system shall provide appointment notifications and reminders through the notification channels supported by the system.

D. Doctor Management

FR-22: The system shall allow doctors to view and manage their professional profiles.

FR-23: The system shall allow doctors to set, update, and manage their appointment availability.

FR-24: The system shall allow doctors to view their scheduled appointments and relevant appointment details.

FR-25: The system shall allow doctors to manage their appointment schedules according to their assigned permissions.

FR-26: The system shall allow doctors to view their previous appointment records.

E. Receptionist Management

FR-27: The system shall allow authorized receptionists to register patients and manage permitted patient information.

FR-28: The system shall allow authorized receptionists to view doctor information and availability.

FR-29: The system shall allow authorized receptionists to create appointments on behalf of patients.

FR-30: The system shall allow authorized receptionists to view, cancel, and reschedule appointments according to clinic policies and assigned permissions.

F. Administrator Management

FR-31: The system shall allow authorized administrators to manage patient information and accounts.

FR-32: The system shall allow authorized administrators to create, update, and manage doctor information and accounts.

FR-33: The system shall allow authorized administrators to manage receptionist accounts.

FR-34: The system shall allow authorized administrators to manage doctor availability.

FR-35: The system shall allow authorized administrators to oversee and manage appointment records.

FR-36: The system shall allow authorized administrators to manage relevant clinic information and system settings.

FR-37: The system shall allow authorized administrators to assign and manage user roles and access permissions.

3.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes and operational characteristics that the Clinic Appointment System must satisfy.

A. Security

NFR-01: The system shall protect user accounts through secure authentication.

NFR-02: The system shall restrict access to information and functions according to users' assigned roles and permissions.

NFR-03: The system shall protect users' personal and sensitive information from unauthorized access.

NFR-04: The system shall securely store user passwords using an appropriate password-hashing mechanism.

NFR-05: The system shall protect information transmitted between users and the system through secure communication protocols.

NFR-06: The system shall prevent unauthorized users from viewing or modifying restricted patient, doctor, and appointment information.

B. Usability

NFR-07: The system shall provide a simple, consistent, and user-friendly interface for all user roles.

NFR-08: The system shall provide clear confirmation messages when an operation is completed successfully.

NFR-09: The system shall display understandable error and validation messages when users provide invalid or incomplete information.

NFR-10: The system shall allow users to navigate the main functions easily and consistently.

C. Performance

NFR-11: The system should respond to normal user actions within an acceptable response time under normal operating conditions.

NFR-12: The system should support multiple users accessing the system simultaneously without significant performance degradation.

D. Reliability and Availability

NFR-13: The system shall maintain accurate and consistent appointment information.

NFR-14: The system shall prevent loss, duplication, or corruption of appointment data during normal operation.

NFR-15: The system shall maintain data consistency when multiple users attempt to access or modify appointment information simultaneously.

NFR-16: The system should provide a backup mechanism to protect important system data against accidental loss.

E. Maintainability

NFR-17: The system shall be designed using a modular structure so that its features can be maintained, modified, and updated easily.

NFR-18: The system shall have sufficient technical documentation to support future maintenance and development.

F. Compatibility and Accessibility

NFR-19: The system should operate correctly on commonly used modern web browsers.

NFR-20: The system should provide a responsive interface that adapts to different screen sizes, including desktop, tablet, and mobile devices.

4. Requirement Refinement Considerations

The following operational policies and implementation details must be clarified and agreed upon by the development team before finalizing the related acceptance criteria.

1. **Patient Registration:** Confirm the required patient registration information and whether any additional account verification is necessary.
2. **Account Management:** Confirm which roles can create, update, and deactivate user accounts.
3. **Patient Information Permissions:** Define which patient profile fields receptionists and administrators can view or edit, particularly sensitive medical information such as blood type and disease type.
4. **Doctor Permissions:** Confirm whether doctors can cancel or reschedule appointments or whether they can only manage their availability and view scheduled appointments.
5. **Cancellation Policy:** Establish the cancellation deadline and the rules for late cancellations.
6. **Rescheduling Policy:** Confirm whether the same cancellation deadline applies to rescheduling appointments.
7. **Appointment Confirmation:** Confirm whether appointment confirmations are displayed on the website, sent by email, or delivered through both methods.
8. **Appointment Reminders:** Establish the reminder schedule and supported notification channels.
9. **Doctor Availability:** Clarify who can modify doctor availability and how existing appointments are handled when a doctor becomes unavailable.
10. **Performance Targets:** Establish measurable response-time and concurrent-user targets for system testing.

5. Scope of the Initial System

The initial version of the Clinic Appointment System focuses on user account management, patient and doctor information management, appointment scheduling, appointment confirmation, cancellation, rescheduling, appointment history, notifications, and role-based access control.

The system is intended to improve the organization and efficiency of clinic appointment operations.

Features such as online payment processing, electronic prescriptions, laboratory management, insurance processing, and comprehensive electronic medical records are not included in the initial project scope unless they are separately approved as additional requirements.

6. Conclusion

The User and System Requirements provide a foundation for the design, development, and testing of the Clinic Appointment System.

The user requirements identify the needs of patients, doctors, receptionists, administrators, and clinic management. The functional requirements translate these needs into specific system operations, while the non-functional requirements define the expected quality, security, usability, performance, and reliability of the system.

These requirements will guide the development team throughout the project and provide a basis for evaluating whether the completed system satisfies the agreed stakeholder expectations.